

SREE PAVAN KOTHAPALLI

MES & Factory Automation Software Engineer | +1 (936) 666-1215 | sree.k1130@gmail.com

PROFESSIONAL SUMMARY

Software Engineer and Data Engineer with 3 years of experience designing, developing, integrating, and supporting scalable enterprise data systems, real-time processing platforms, cloud-native architectures, and automated workflows. Strong technical expertise in Python, SQL, Apache Kafka, Azure, AWS, Databricks, PySpark, Docker, Kubernetes, and CI/CD automation. Experienced in enterprise system integration, event-driven architectures, operational data processing, data validation, application monitoring, troubleshooting, and system reliability. Hands-on academic project experience in Industrial IoT, sensor data ingestion, equipment health monitoring, and predictive maintenance analytics. Skilled in implementing secure, scalable, and reliable software solutions with transferable capabilities relevant to Manufacturing Execution Systems (MES), factory automation, manufacturing data integration, and AI-enabled industrial applications.

TECHNICAL SKILLS

- MES & Factory Automation:** MES Configuration & Support, CIM/EAP Integration, Equipment Interfaces, Manufacturing Master Data, Traceability, Quality Data Integration, UAT, Go-Live & Post Go-Live Support
- Programming Languages:** Python, SQL, Scala, Bash, T-SQL, PowerShell
- Real-Time Integration & Automation:** Apache Kafka, Kafka Connect, Kafka Streams, Schema Registry, Azure Event Hubs, Azure Stream Analytics, Event-Driven Architecture, Enterprise Application Integration, Automated Data Processing, Real-Time Data Ingestion
- Industrial IoT & Equipment Analytics:** Azure IoT Hub, IoT Sensor Data Processing, Equipment Health Monitoring, Predictive Maintenance Analytics, Industrial Data Analytics, Power BI Monitoring Dashboards
- Data Engineering & Processing:** Apache Spark, PySpark, Databricks, Delta Lake, Spark SQL, Apache Airflow, Azure Data Factory, Hadoop, Hive, HDFS, Azure Synapse Analytics
- Cloud Platforms:** Microsoft Azure, AWS S3, AWS EMR, AWS Glue, Azure Blob Storage, Azure Data Lake Gen2, Azure Functions
- Databases:** PostgreSQL, MySQL, SQL Server, Oracle, Snowflake, Azure SQL Database
- Data Quality & Reliability:** Data Validation, Data Integrity, Data Quality Controls, Great Expectations, Error Handling, Data Consistency, Data Transformation, Performance Optimization
- DevOps & Deployment:** Git, Jenkins, Azure DevOps, Docker, Kubernetes, CI/CD Pipelines, Linux, Bash Automation
- Monitoring & Troubleshooting:** Prometheus, Grafana, Azure Monitor, System Monitoring, Alerting, Performance Troubleshooting, Operational Dashboards
- AI & Analytics Tools:** MLflow, Databricks, Power BI, Tableau, Predictive Maintenance Analytics
- Documentation:** Confluence, Draw.io, Markdown, Technical Documentation, Data Flow Diagrams

PROFESSIONAL EXPERIENCE

Data Engineer

Jun 2024 - Present

Alephys LLC

Houston, Texas

- Designed and implemented scalable, real-time enterprise data integration solutions using Apache Kafka, Kafka Connect, and Schema Registry to enable reliable communication between distributed applications and data platforms, handling 1M+ events per minute with sub-second latency.
- Developed automated data processing workflows using Python, SQL, Databricks, and Delta Lake to support high-volume operational data ingestion, transformation, and analytics.
- Configured Kafka Connect and Schema Registry to maintain consistent data structures and enable reliable data exchange across heterogeneous enterprise systems.
- Integrated Azure Data Factory with Kafka streaming pipelines to automate data movement between on-premises and cloud-based platforms.
- Implemented data consistency, security, and validation controls to improve data integrity and reliability across distributed processing systems.
- Built real-time data processing applications using Databricks and Delta Lake, supporting large-scale streaming and analytical workloads of 50+ TB of data monthly.
- Optimized Spark SQL queries and data models to improve processing performance (50% improvement) and reduce infrastructure overhead.
- Developed and maintained CI/CD pipelines using Jenkins and Git to automate application deployments and support reliable software releases.
- Implemented access controls and encryption mechanisms to secure distributed data integration infrastructure.
- Supported scalable, event-driven architectures capable of handling high-throughput operational data with low processing latency.

MES & Manufacturing Systems

- Developed, configured, tested, and supported Manufacturing Execution System (MES) applications to enable automated production execution, process monitoring, and manufacturing operations.
- Translated manufacturing business requirements into MES configurations, system enhancements, technical specifications, and automated workflows.
- Integrated MES with Computer Integrated Manufacturing (CIM), Equipment Automation Programs (EAP), and manufacturing data platforms to enable seamless production data exchange.
- Configured and maintained manufacturing master data, including process flows, equipment definitions, production recipes, material information, and operational parameters.
- Supported MES functionality for work-order execution, production tracking, equipment status monitoring, and manufacturing process traceability.

Equipment Integration & Factory Automation

- Developed and supported equipment integration interfaces to enable automated exchange of equipment events, process parameters, and production status information.
- Collaborated with equipment vendors and automation engineers to configure, test, and troubleshoot manufacturing equipment interfaces.
- Integrated factory automation systems with enterprise data platforms to support real-time equipment monitoring and manufacturing analytics.
- Supported automated equipment data collection and event-driven workflows to reduce manual intervention in factory operations.

Quality & Manufacturing Data

- Integrated manufacturing quality data with production execution systems to support quality monitoring, validation, and traceability.
- Implemented data validation and reconciliation controls to improve the accuracy and consistency of manufacturing process data.
- Supported manufacturing data models covering materials, recipes, process steps, equipment, and production events.

Testing, UAT & Go-Live

- Participated in functional testing, integration testing, User Acceptance Testing (UAT), and defect resolution for MES and factory automation applications.
- Supported system deployment, manufacturing go-live readiness, post-go-live stabilization, and production issue resolution.
- Investigated application failures, equipment interface errors, and manufacturing data inconsistencies to minimize operational disruption.
- Prepared test cases, deployment documentation, troubleshooting guides, and standard operating procedures.

Collaboration & AI-Enabled Manufacturing

- Collaborated with manufacturing, quality, IT, automation, and vendor teams to gather requirements and deliver integrated factory software solutions.
- Developed AI-enabled manufacturing analytics workflows to identify equipment anomalies and support predictive maintenance.
- Supported implementation of scalable factory systems in a greenfield manufacturing environment.

Environment: Kafka, Kafka Connect, Schema Registry, Databricks, Spark, Delta Lake, Airflow, ADF, AWS S3/EMR/Glue, Azure Data Lake Gen2, Python, SQL, Bash, Scala, Jenkins, Git, Docker, MLflow, Great Expectations, SQL Server, PostgreSQL, Snowflake, Prometheus, Grafana, Azure Monitor

Data Engineer (Intern)

Jan 2024 – May 2024

Alephys LLC

Houston, Texas

- Supported senior engineers delivering enterprise data integration solutions for Fortune 500 clients.
- Built ETL pipelines (Spark, Hive) ingesting 500GB+ daily from diverse sources; partitioned Hive tables cut query times from minutes to seconds.
- Migrated 50+ tables from on-premises Hadoop to Azure Synapse with complex transformations while preserving data integrity.
- Configured Kafka producers/consumers for real-time streaming between on-premises and cloud systems with robust error handling.
- Developed Power BI dashboards connecting to Databricks and Hadoop data sources for business reporting.

Data Engineer

May 2021 – May 2022

Birlasoft

Chennai, India

- Developed automated data processing and operational workflows using Python, SQL, Linux, and Bash scripting to improve system reliability and reduce manual intervention.
- Built real-time data processing frameworks using Azure Stream Analytics and Azure Event Hubs to enable continuous ingestion and processing of event-driven data.
- Designed and implemented scalable Azure SQL data pipelines supporting enterprise operational reporting and analytics, processing 10TB+ monthly with 99.5% reliability.
- Automated data validation, backup, and system maintenance activities using Linux shell scripting.
- Configured monitoring and alerting mechanisms to identify processing failures and support timely troubleshooting of operational issues.
- Administered Linux (Oracle Linux) environments supporting mission-critical data processing applications, including performance tuning and security hardening.

- Managed Hadoop clusters for distributed processing of large financial datasets.
- Developed Power BI dashboards to visualize operational data and provide actionable insights to business stakeholders.
- Designed and optimized analytical data models in Azure Synapse, improving reporting performance by 45%.

SELECTED PROJECTS

Azure-Based Industrial IoT and Equipment Monitoring Pipeline — Master's Project

- Designed and developed an end-to-end Industrial IoT data processing solution using Azure IoT Hub, Azure Stream Analytics, and Azure Data Lake to ingest and process real-time sensor data from connected devices.
- Implemented event-driven data processing workflows to collect, transform, and store sensor-generated information for operational analytics.
- Developed Azure Functions to perform data cleansing and transformation before persisting processed sensor data in Azure Blob Storage.
- Integrated Power BI dashboards to visualize device health metrics and provide insights supporting predictive maintenance in industrial applications.
- Applied cloud-native, serverless architecture principles to develop scalable and reliable IoT data processing workflows.
- Demonstrated real-time monitoring and analytics capabilities relevant to industrial equipment performance and automated manufacturing environments.

Kafka-Driven Real-Time Data Pipeline

- Deployed a fault-tolerant, high-throughput Kafka pipeline on Cloudera with Kafka Connect source integration and KSQL DB real-time analytics.

On-Premises to Azure Data Migration

- Migrated large datasets from SQL Server to Azure Data Lake Gen2 with Databricks ETL and Synapse analytics, ensuring consistency and integrity; added Kafka streaming to cut latency.

EDUCATION

Master of Science in Computer Science, Univ. of North Texas, Denton, TX

Aug 2022 – May 2024

CERTIFICATIONS

- Confluent Certified Administrator for Apache Kafka
- Databricks Certified Data Engineer Professional